Project: Worked examples
Title: Fillet weld (EN 1993-1-8 simplified method)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A fillet weld carrying forces parallel to, transverse to and normal to its axis, by the simplified method: the resultant over the throat area against the design shear strength of the weld, with the required throat and leg.

F = √(F_∥² + F_⊥² + F_n²); f_vw,d = f_u/(√3·β_w·γ_M2); τ_w = F/(a·L); F_w,Rd = f_vw,d·a·L; u = F/F_w,Rd; a_req = max(F/(f_vw,d·L), 3 mm); leg = √2·a.

Ref: BS EN 1993-1-8 4.5.3.3 — Simplified method for design resistance of fillet weld
Ref: BS EN 1993-1-8 4.5.2(2) — Effective throat thickness: minimum 3 mm
Ref: BS EN 1993-1-8 4.5.1(2) — Effective length: welds shorter than 30 mm or 6 throats not to carry load
Ref: BS EN 1993-1-8 4.11 — Long joints

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_par	Force parallel to the weld axis	150	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_tr	Force transverse to the weld axis, in the plane of the parts	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_n	Force normal to the plane of the parts	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
L	Effective weld length	200	mm	drawing: DRG-WD-701 rev A	30 … 10000 mm
a	Throat thickness	5	mm	drawing: DRG-WD-701 rev A	3 … 50 mm
f_u	Ultimate strength of the weaker part joined	490	MPa	spec: SPEC-STR-02 design criteria	300 … 1000 MPa
beta_w	Correlation factor β_w (0.8 S235, 0.85 S275, 0.9 S355, 1.0 S420/S460)	0.9		spec: SPEC-STR-02 design criteria	0.8 … 1
gamma_M2	Partial factor γ_M2	1.25		spec: SPEC-STR-02 design criteria	1 … 2

3. Assumptions

A1. The resultant force is shared uniformly along the effective length. (Simplified method.)
A2. Equal-leg fillet: leg = √2 × throat. (Geometry.)

4. Working

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[9] F = sqrt(F_par^2 + F_tr^2 + F_n^2)
= sqrt((150 kN)^2 + (0 kN)^2 + (0 kN)^2)
= 150 kN
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Resultant force on the weld

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[10] f_vw_d = f_u / (sqrt(3) * beta_w * gamma_M2)
= 490 MPa / (sqrt(3) × 0.9 × 1.25)
= 251.47 MPa
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BS EN 1993-1-8 4.5.3.3

Design shear strength of the weld

[11] tau_w = F / (a * L)
= 150 kN / (5 mm × 200 mm)
= 150 MPa

Stress on the throat

[12] F_w_Rd = f_vw_d * a * L — BS EN 1993-1-8 4.5.3.3
= 251.47 MPa × 5 mm × 200 mm
= 251.47 kN

Design resistance of the weld

[13] u = F / F_w_Rd
= 150 kN / 251.47 kN
= 0.5965

Utilisation

[14] a_calc = F / (f_vw_d * L)
= 150 kN / (251.47 MPa × 200 mm)
= 2.9825 mm

Throat needed for the load

[15] a_req = max(a_calc, 3 mm) — BS EN 1993-1-8 4.5.2(2)
= max(2.9825 mm, 3 mm)
= 3 mm

Required throat (not less than 3 mm)

[16] s_leg = sqrt(2) * a
= sqrt(2) × 5 mm
= 7.0711 mm

Leg length of the throat provided (equal legs)

5. Results

SYMBOL	VALUE	DESCRIPTION
F	150 kN	Resultant force
f_vw_d	251.47 MPa	Design shear strength
tau_w	150 MPa	Throat stress
F_w_Rd	251.47 kN	Weld resistance
u	0.5965	Utilisation
a_calc	2.9825 mm	Throat needed
a_req	3 mm	Required throat
s_leg	7.0711 mm	Leg length provided

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Weld resistance	u = 0.5965	u ≤ 1	0.596	BS EN 1993-1-8 4.5.3.3

7. Applicability

No warnings.

8. Verification

Template struct.fillet-weld@1.0.0 Content hash 74b4c9ab443f75d4 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0